RAJALAKSHMI ENGINEERING COLLEGE
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AD23532 – PRINCIPLES OF DATA SCIENCE
EXPERIMENT 2: DATA DISCOVERY AND PREPARATION

QUESTION:

A telecommunications company has observed an increase in customer churn, leading to revenue loss. Analyze the Telco Customer Churn dataset to discover meaningful insights, identify data quality issues, detect outliers, visualize customer behavior, and prepare the dataset for predictive analytics.

AIM:

To perform data discovery and exploratory data analysis on the Telco Customer Churn dataset by analyzing customer distributions, visualizing numerical features, detecting outliers using the IQR method, identifying relationships between variables, answering analytical questions, splitting the dataset into training and testing sets, and saving the preprocessed dataset as a CSV file.

ALGORITHM:

1. Import the required Python libraries: Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.
2. Load the Telco Customer Churn dataset into a Pandas DataFrame and display its basic information and summary statistics.
3. Analyze the distributions of Customer Churn, Gender, Contract Type, and Payment Method using count plots.
4. Visualize MonthlyCharges, TotalCharges, and Tenure using histograms and box plots.
5. Detect outliers in numerical features using the Interquartile Range (IQR) method.
6. Create count plots, box plots, histograms, and scatter plots to analyze relationships between Contract Type, Internet Service, Monthly Charges, Tenure, and Churn.
7. Analyze churn patterns, frequently used payment methods among churned customers, average tenure, monthly charges, and customer groups with higher churn.
8. Split the dataset into 80% training data and 20% testing data.
9. Save the prepared dataset as CustomerChurn_Preprocessed.csv.

PROGRAM & OUTPUT:

```
# Name: Shyam M (241801267)
# Section 1: Import Libraries and Load Dataset

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split

df = pd.read_csv("/content/WA_Fn-UseC_-Telco-Customer-Churn.csv")

print("Dataset Shape:", df.shape)
print(df.head())
print(df.info())
print(df.describe(include="all"))
```

```
Dataset Shape: (7043, 21)
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes

              PaymentMethod MonthlyCharges  TotalCharges Churn
0          Electronic check          29.85         29.85    No
1              Mailed check          56.95        1889.5    No
2              Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4          Electronic check          70.70        151.65   Yes

[5 rows x 21 columns]
```

```
[5 rows x 21 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
None
       customerID gender  SeniorCitizen Partner Dependents       tenure  \
count        7043   7043    7043.000000    7043        7043  7043.000000
unique       7043      2            NaN       2           2          NaN
top     3186-AJIEK   Male            NaN      No          No          NaN
freq            1   3555            NaN    3641        4933          NaN
mean          NaN    NaN       0.162147     NaN         NaN    32.371149
std           NaN    NaN       0.368612     NaN         NaN    24.559481
min           NaN    NaN       0.000000     NaN         NaN     0.000000
25%           NaN    NaN       0.000000     NaN         NaN     9.000000
50%           NaN    NaN       0.000000     NaN         NaN    29.000000
75%           NaN    NaN       0.000000     NaN         NaN    55.000000
max           NaN    NaN       1.000000     NaN         NaN    72.000000

       PhoneService MultipleLines InternetService OnlineSecurity  ... \
count          7043          7043            7043           7043  ...
unique            2             3               3              3  ...
top             Yes            No     Fiber optic             No  ...
freq           6361          3390            3096           3498  ...
mean            NaN           NaN             NaN            NaN  ...
std             NaN           NaN             NaN            NaN  ...
min             NaN           NaN             NaN            NaN  ...
25%             NaN           NaN             NaN            NaN  ...
50%             NaN           NaN             NaN            NaN  ...
75%             NaN           NaN             NaN            NaN  ...
max             NaN           NaN             NaN            NaN  ...

       DeviceProtection TechSupport StreamingTV StreamingMovies  \
count              7043        7043        7043            7043
```

```
       DeviceProtection TechSupport StreamingTV StreamingMovies  \
count              7043        7043        7043            7043
unique                3           3           3               3
top                  No          No          No              No
freq               3095        3473        2810            2785
mean                NaN         NaN         NaN             NaN
std                 NaN         NaN         NaN             NaN
min                 NaN         NaN         NaN             NaN
25%                 NaN         NaN         NaN             NaN
50%                 NaN         NaN         NaN             NaN
75%                 NaN         NaN         NaN             NaN
max                 NaN         NaN         NaN             NaN

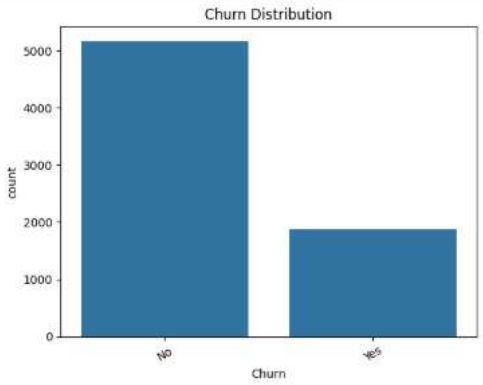
              Contract PaperlessBilling PaymentMethod MonthlyCharges  \
count             7043             7043          7043   7043.000000
unique               3                2             4           NaN
top     Month-to-month              Yes  Electronic check           NaN
freq              3875             4171          2365           NaN
mean               NaN              NaN           NaN     64.761692
std                NaN              NaN           NaN     30.090047
min                NaN              NaN           NaN     18.250000
25%                NaN              NaN           NaN     35.500000
50%                NaN              NaN           NaN     70.350000
75%                NaN              NaN           NaN     89.850000
max                NaN              NaN           NaN    118.750000

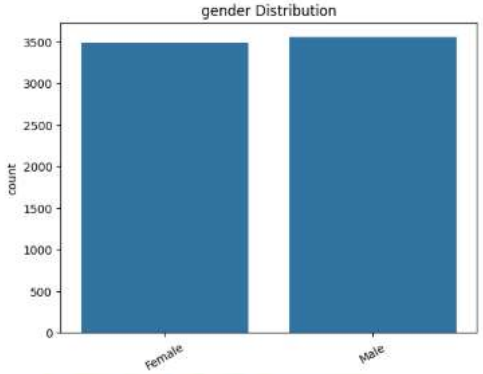
       TotalCharges Churn
count          7043  7043
unique         6531     2
top                   No
freq             11  5174
mean            NaN   NaN
std             NaN   NaN
min             NaN   NaN
25%             NaN   NaN
50%             NaN   NaN
75%             NaN   NaN
max             NaN   NaN

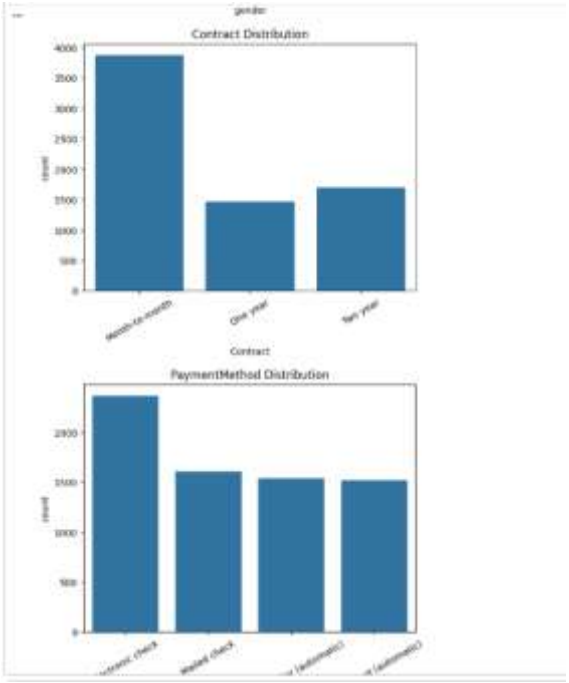
[11 rows x 21 columns]
```

```
# Name: Shyam M (241801267)
# Section 2: Analyze Categorical Distributions

for col in ["Churn", "gender", "Contract", "PaymentMethod"]:
    if col in df.columns:
        sns.countplot(data=df, x=col)
        plt.title(col + " Distribution")
        plt.xticks(rotation=30)
        plt.show()
```





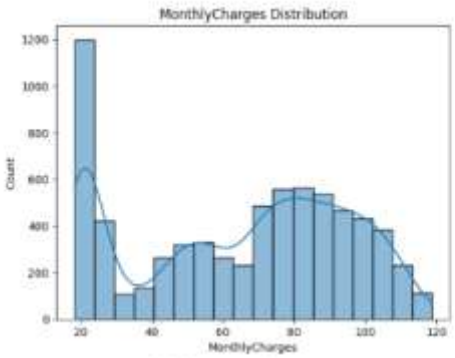


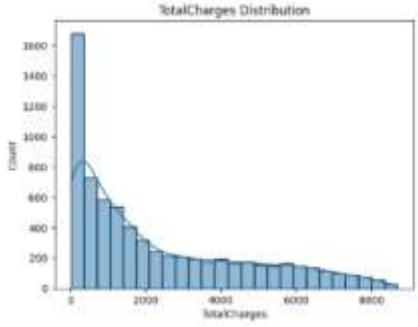

```
# Name: Shyam M (241801267)
# Section 3: Visualize Numerical Features

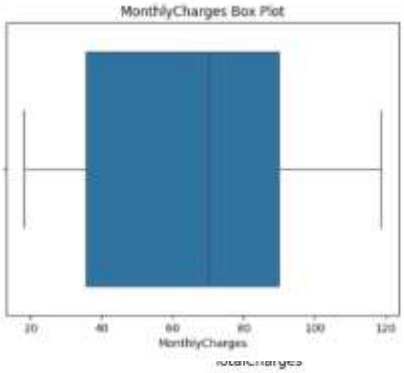
for col in ["MonthlyCharges", "TotalCharges", "tenure"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")

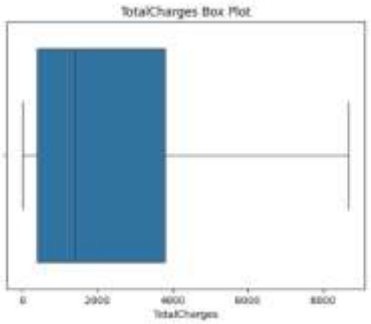
    sns.histplot(df[col].dropna(), kde=True)
    plt.title(col + " Distribution")
    plt.show()

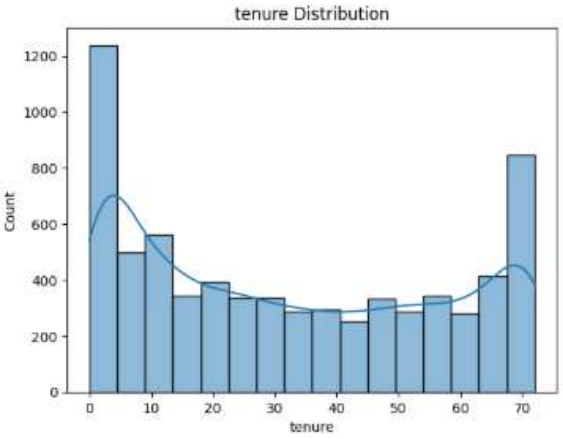
    sns.boxplot(x=df[col])
    plt.title(col + " Box Plot")
    plt.show()
```

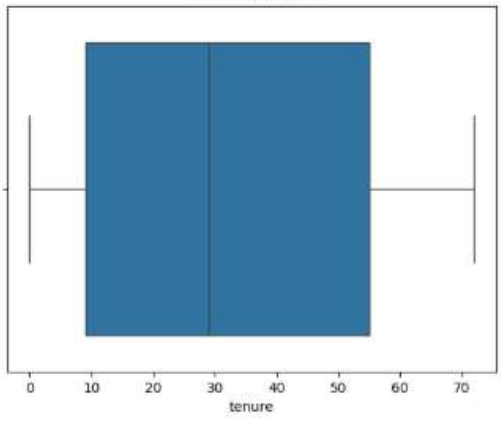

MonthlyCharges Distribution
Count
MonthlyCharges


TotalCharges Distribution
Count
TotalCharges


MonthlyCharges Box Plot
MonthlyCharges


TotalCharges Box Plot
TotalCharges


tenure Distribution
Count
tenure


tenure Box Plot
tenure

```
# Name: Shyam M (241801267)
# Section 4: Detect Outliers Using IQR

for col in ["MonthlyCharges", "TotalCharges", "tenure"]:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    count = ((df[col] < lower) | (df[col] > upper)).sum()
    print(col, "Outliers:", count)
```

```
MonthlyCharges Outliers: 0
TotalCharges Outliers: 0
tenure Outliers: 0
```

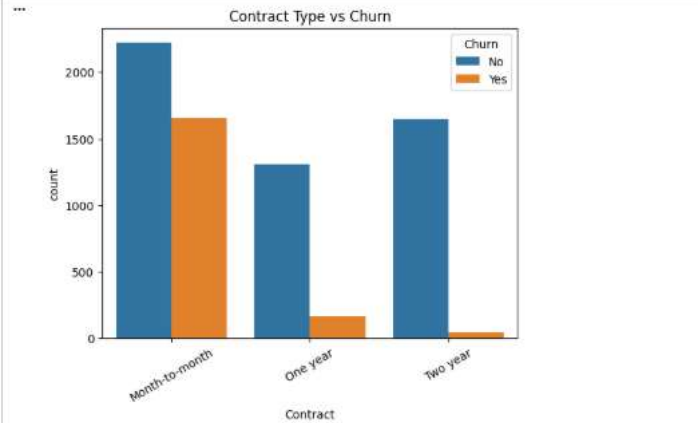
```
# Name: Shyam M (241801267)
# Section 5: Visualize Relationships

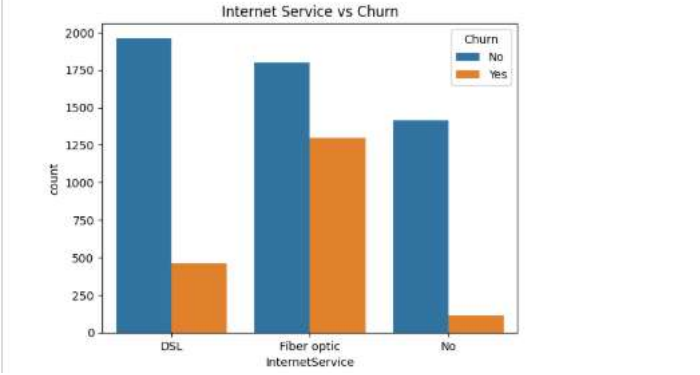
sns.countplot(data=df, x="Contract", hue="Churn")
plt.title("Contract Type vs Churn")
plt.xticks(rotation=30)
plt.show()

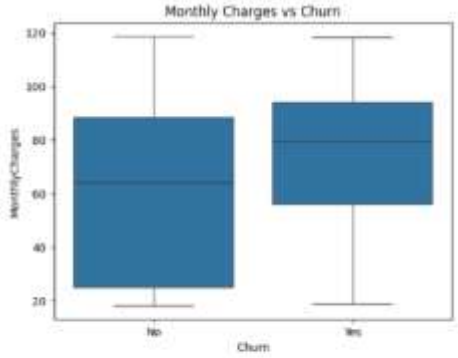
sns.countplot(data=df, x="InternetService", hue="Churn")
plt.title("Internet Service vs Churn")
plt.show()

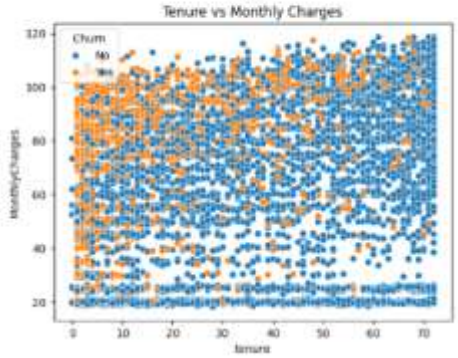
sns.boxplot(data=df, x="Churn", y="MonthlyCharges")
plt.title("Monthly Charges vs Churn")
plt.show()

sns.scatterplot(data=df, x="tenure",
                y="MonthlyCharges", hue="Churn")
plt.title("Tenure vs Monthly Charges")
plt.show()
```










```
# Name: Shyam M (241801267)
# Section 6: Analyze Customer Churn

print("Churn by Contract:\n",
      df.groupby("Contract")["Churn"].value_counts())

print("Payment Methods of Churned Customers:\n",
      df[df["Churn"] == "Yes"]["PaymentMethod"].value_counts())

print("Average Tenure by Churn:\n",
      df.groupby("Churn")["tenure"].mean())

print("Average Monthly Charges by Churn:\n",
      df.groupby("Churn")["MonthlyCharges"].mean())

print("Churn by Internet Service:\n",
      df.groupby("InternetService")["Churn"].value_counts())
```

```
Churn by Contract:
 Contract        Churn
Month-to-month  No       2220
                Yes      1655
One year        No       1307
                Yes       166
Two year        No       1647
                Yes        48
Name: count, dtype: int64
Payment Methods of Churned Customers:
 PaymentMethod
Electronic check             1071
Mailed check                  308
Bank transfer (automatic)     258
Credit card (automatic)       232
Name: count, dtype: int64
Average Tenure by Churn:
 Churn
No     37.569965
Yes    17.979133
Name: tenure, dtype: float64
Average Monthly Charges by Churn:
 Churn
No     61.265124
Yes    74.441332
Name: MonthlyCharges, dtype: float64
Churn by Internet Service:
 InternetService  Churn
DSL              No       1962
                 Yes       459
Fiber optic      No       1799
                 Yes      1297
No               No       1413
                 Yes       113
Name: count, dtype: int64
```

```
# Name: Shyam M (241801267)
# Section 7: Split Dataset into Training and Testing Sets

X = df.drop(columns=["Churn", "customerID"], errors="ignore")
y = df["Churn"]

X = pd.get_dummies(X, drop_first=True, dtype=int)
X = X.fillna(X.median(numeric_only=True)).fillna(0)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42,
    stratify=y
)

print("Training set:", X_train.shape)
print("Testing set:", X_test.shape)
```

```
Training set: (5634, 30)
Testing set: (1409, 30)
```

```
# Name: Shyam M (241801267)
# Section 8: Save Prepared Dataset

df.to_csv("CustomerChurn_Preprocessed.csv", index=False)
print("Dataset saved successfully.")
```

```
Dataset saved successfully.
```

RESULT:

Thus, data discovery and exploratory data analysis were performed successfully on the Telco Customer Churn dataset. Customer distributions were analyzed, numerical outliers were identified using the IQR method, and visualizations were generated to study relationships between customer features and churn. The dataset was split into 80% training data and 20% testing data, and the prepared dataset was saved as CustomerChurn_Preprocessed.csv.